# GT: 7.1 final on gpt-5.4-mini against gpt-5.4

Generated 2026-10-04 10:15 EDT. Sources: /Users/elenabaron/MITxPro/RAGandContextEng/capstone/rag_and_context_engineering/wikipedia_retrieval_engine/Capstone Checkpoint 7.1/cost_efficiency/wikipedia_GT

| Question set | Configurations | Runs | Question-runs | Baseline | Highest quality | Lowest list cost | Fastest |
|---|---|---|---|---|---|---|---|
| GT test set | 3 | 7 | 133 | 7.1 final | 7.1 final, gpt-5.4 (0.868) | 6.1 survivor ($0.0485) | 6.1 survivor (13.1 s) |

## How to read this report

**Quality.** pass = 1, partial = 0.5, fail = 0, averaged over question-runs (RAGAS judge).

**List cost.** every input token at the full price: the comparison of record. Billed cost charges cached input at the cached rate, so it falls when runs repeat the same questions.

**Seconds.** agent graph plus answer per question, judge excluded. Provider speed varies by hour; §6 shows how much.

**Dominated.** another configuration is at least as good and no more expensive (or slower), and better on one of the two.

**Paired.** each question's mean over its runs, configuration minus baseline; p from the Wilcoxon signed-rank test over the questions that changed.

**Paths.** Table: the TOC review flagged a table. Text, escalated: an assessment asked for more evidence. Text, ready: neither.

## Loader warnings

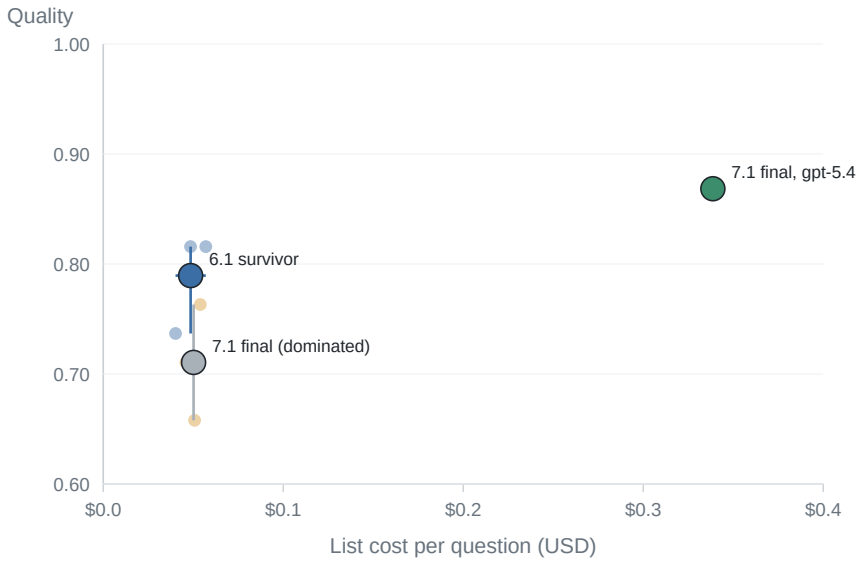
*- adoring_hopper: list cost repriced from its calls with the current price table ($4.9619 recorded -> $6.4348, +29.7%).*

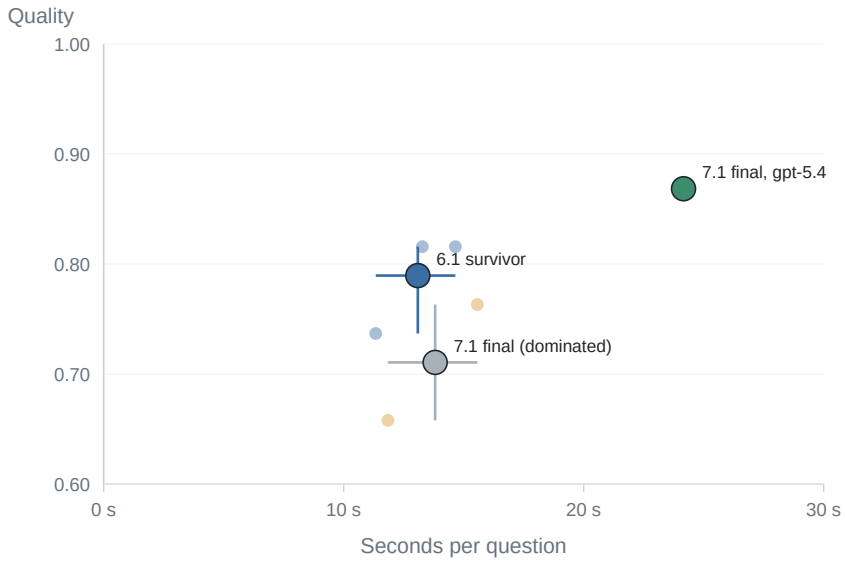
# GT test set

3 configuration(s), 7 run(s), 133 question-runs. Baseline: 7.1 final (requested).

## 1. Frontier: quality against cost and time

| Configuration | Runs (with telemetry) | Quality (run range) | Pass / partial / fail | List $/q (run range) | Billed $/q | Cached | Seconds mean / median | Calls/q | Input tokens/q | List $ per quality point | Dominated on cost by | Dominated on time by |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 6.1 survivor | 3 (3) | 0.789 (0.737-0.816) | 40/10/7 | $0.0485 ($0.0402-$0.0570) | $0.0350 | 35% | 13.1 / 11.8 | 6.77 | 56.8K | $0.0615 | - | - |
| 7.1 final | 3 (3) | 0.711 (0.658-0.763) | 37/7/13 | $0.0502 ($0.0459-$0.0539) | $0.0348 | 38% | 13.8 / 12.6 | 6.84 | 59.3K | $0.0706 | 6.1 survivor | 6.1 survivor |
| 7.1 final, gpt-5.4 | 1 (1) | 0.868 (0.868) | 16/1/2 | $0.3387 ($0.3387) | $0.2787 | 20% | 24.2 / 22.3 | 7.11 | 95.8K | $0.3900 | - | - |



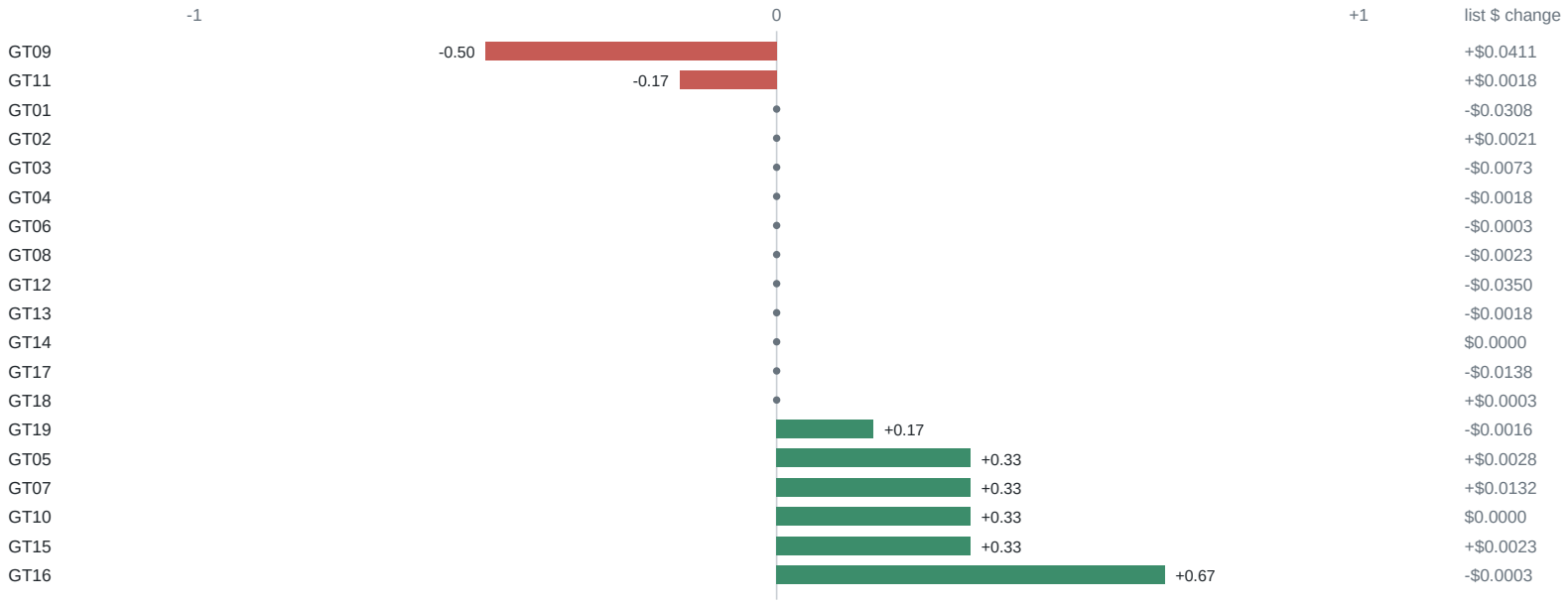

*Small dots are runs; a large dot is the configuration mean, with lines across its run range. Gray: dominated.*

## 2. Paired against the baseline (7.1 final)

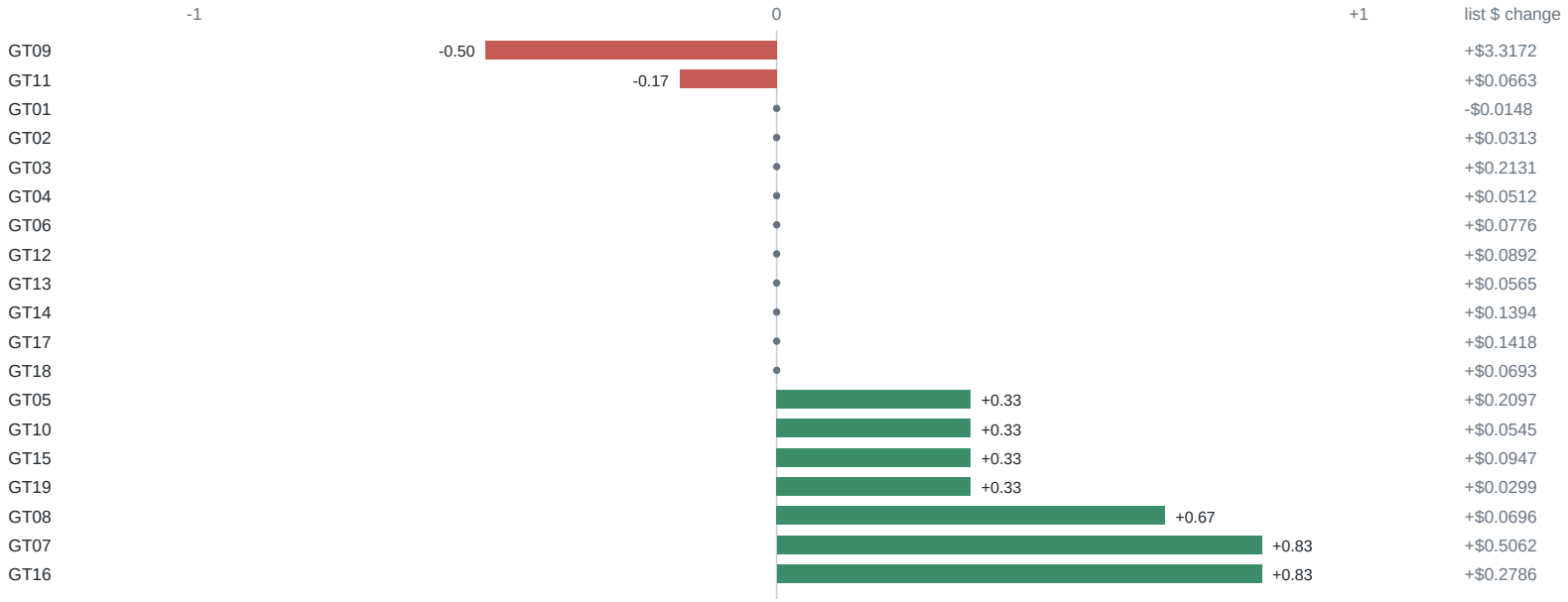
| Configuration | Questions | Quality | Δ quality | Better / worse / same | p (quality) | Δ list $/q | p (cost) | Δ seconds | p (time) | Trade-off |
|---|---|---|---|---|---|---|---|---|---|---|
| 6.1 survivor | 19 | 0.711 -> 0.789 | +0.079 | 6 / 2 / 11 | 0.188 | -$0.0016 (-3%) | 0.679 | -0.7 | 0.061 | dominates the baseline: at least as good, no more expensive |
| 7.1 final, gpt-5.4 | 19 | 0.711 -> 0.868 | +0.158 | 7 / 2 / 10 | 0.074 | +$0.2885 (+575%) | <0.001 | +10.3 | <0.001 | pays $0.1827 per question for each +0.10 quality |

*Per-question change in quality, worst first (right column: change in list cost):*

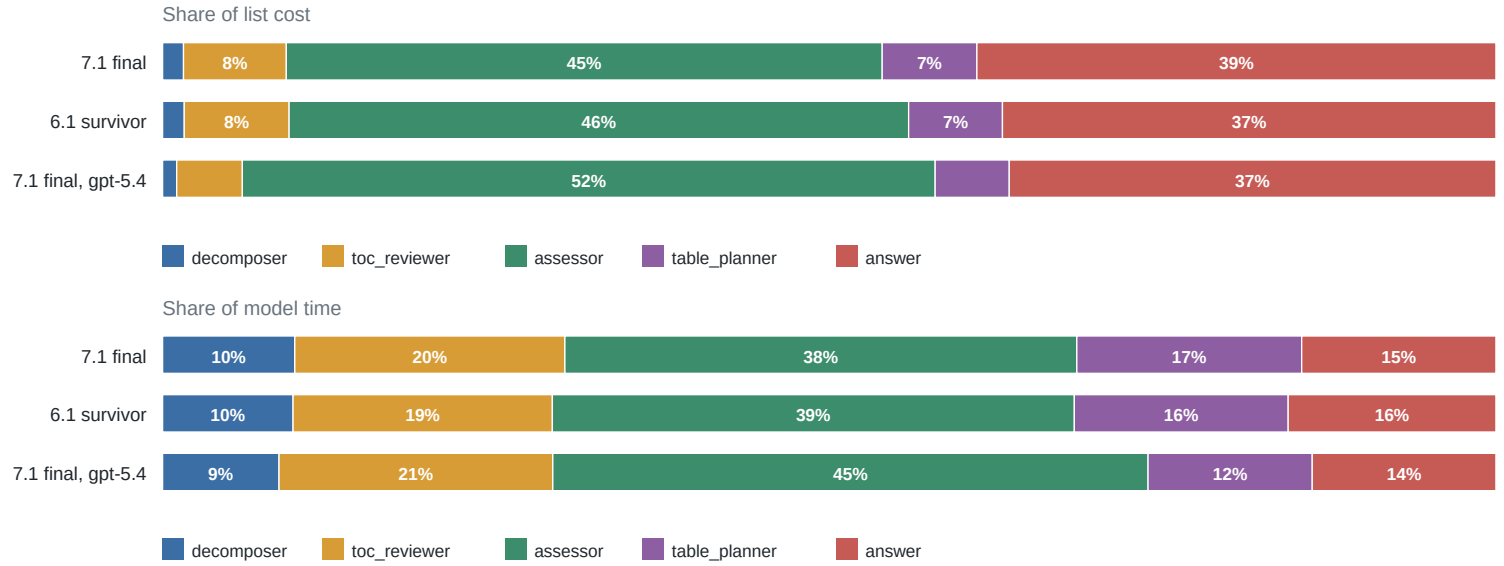
6.1 survivor against 7.1 final




7.1 final, gpt-5.4 against 7.1 final

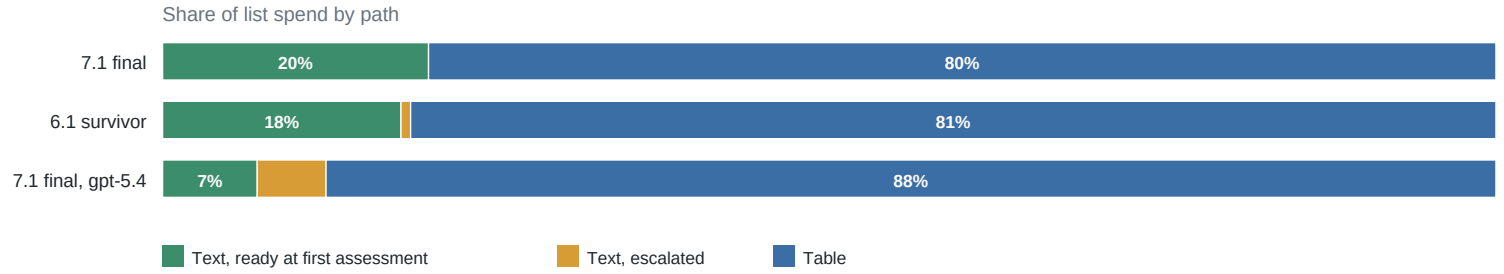

-1
0
+1
list $ change
GT09
-0.50
+$3.3172
GT11
-0.17
+$0.0663
GT01
-$0.0148
GT02
+$0.0313
GT03
+$0.2131
GT04
+$0.0512
GT06
+$0.0776
GT12
+$0.0892
GT13
+$0.0565
GT14
+$0.1394
GT17
+$0.1418
GT18
+$0.0693
GT05
+0.33
+$0.2097
GT10
+0.33
+$0.0545
GT15
+0.33
+$0.0947
GT19
+0.33
+$0.0299
GT08
+0.67
+$0.0696
GT07
+0.83
+$0.5062
GT16
+0.83
+$0.2786

## 3. Where the cost goes: by role




| Configuration | Role | Calls/q | Input tokens/q | Output tokens/q | Cached | List $/q | Share of list | Model s/q | Share of time | Median s/call |
|---|---|---|---|---|---|---|---|---|---|---|
| 7.1 final | decomposer | 1.00 | 0.4K | 104 | 0% | $0.0008 | 2% | 1.1 | 10% | 1.04 |
| 7.1 final | toc_reviewer | 1.00 | 3.0K | 361 | 82% | $0.0039 | 8% | 2.3 | 20% | 2.20 |
| 7.1 final | assessor | 2.02 | 26.9K | 505 | 34% | $0.0224 | 45% | 4.3 | 38% | 1.89 |
| 7.1 final | table_planner | 1.82 | 3.9K | 140 | 47% | $0.0036 | 7% | 1.9 | 17% | 1.00 |
| 7.1 final | answer | 1.00 | 25.1K | 149 | 37% | $0.0195 | 39% | 1.6 | 15% | 1.52 |
| 6.1 survivor | decomposer | 1.00 | 0.4K | 105 | 0% | $0.0008 | 2% | 1.0 | 10% | 1.02 |
| 6.1 survivor | toc_reviewer | 1.00 | 3.0K | 353 | 55% | $0.0038 | 8% | 2.1 | 19% | 1.99 |
| 6.1 survivor | assessor | 2.09 | 26.9K | 534 | 38% | $0.0226 | 46% | 4.2 | 39% | 1.85 |
| 6.1 survivor | table_planner | 1.68 | 3.7K | 137 | 38% | $0.0034 | 7% | 1.7 | 16% | 0.95 |
| 6.1 survivor | answer | 1.00 | 22.8K | 184 | 30% | $0.0180 | 37% | 1.7 | 16% | 1.42 |
| 7.1 final, gpt-5.4 | decomposer | 1.00 | 0.4K | 114 | 0% | $0.0028 | 1% | 1.9 | 9% | 1.80 |
| 7.1 final, gpt-5.4 | toc_reviewer | 1.00 | 3.0K | 351 | 0% | $0.0129 | 5% | 4.5 | 21% | 4.42 |
| 7.1 final, gpt-5.4 | assessor | 2.16 | 50.2K | 683 | 35% | $0.1357 | 52% | 9.7 | 45% | 4.38 |
| 7.1 final, gpt-5.4 | table_planner | 1.95 | 5.0K | 134 | 19% | $0.0145 | 6% | 2.7 | 12% | 1.34 |
| 7.1 final, gpt-5.4 | answer | 1.00 | 37.1K | 168 | 0% | $0.0953 | 37% | 3.0 | 14% | 2.35 |

## 4. By path: text ready, text escalated, table




| Configuration | Path | Question-runs (share) | Quality | List $/q | Billed $/q | Seconds/q | Share of list spend |
|---|---|---|---|---|---|---|---|
| 7.1 final | Text, ready at first assessment | 15 (26%) | 0.833 | $0.0380 | $0.0191 | 9.3 | 20% |
| 7.1 final | Table | 42 (74%) | 0.667 | $0.0545 | $0.0404 | 15.4 | 80% |
| 6.1 survivor | Text, ready at first assessment | 13 (23%) | 0.962 | $0.0381 | $0.0203 | 8.9 | 18% |
| 6.1 survivor | Text, escalated | 1 (2%) | 1.000 | $0.0200 | $0.0200 | 8.2 | 1% |
| 6.1 survivor | Table | 43 (75%) | 0.733 | $0.0524 | $0.0398 | 14.5 | 81% |
| 7.1 final, gpt-5.4 | Text, ready at first assessment | 4 (21%) | 1.000 | $0.1142 | $0.1142 | 13.5 | 7% |
| 7.1 final, gpt-5.4 | Text, escalated | 1 (5%) | 1.000 | $0.3322 | $0.3322 | 22.3 | 5% |
| 7.1 final, gpt-5.4 | Table | 14 (74%) | 0.821 | $0.4033 | $0.3219 | 27.3 | 88% |

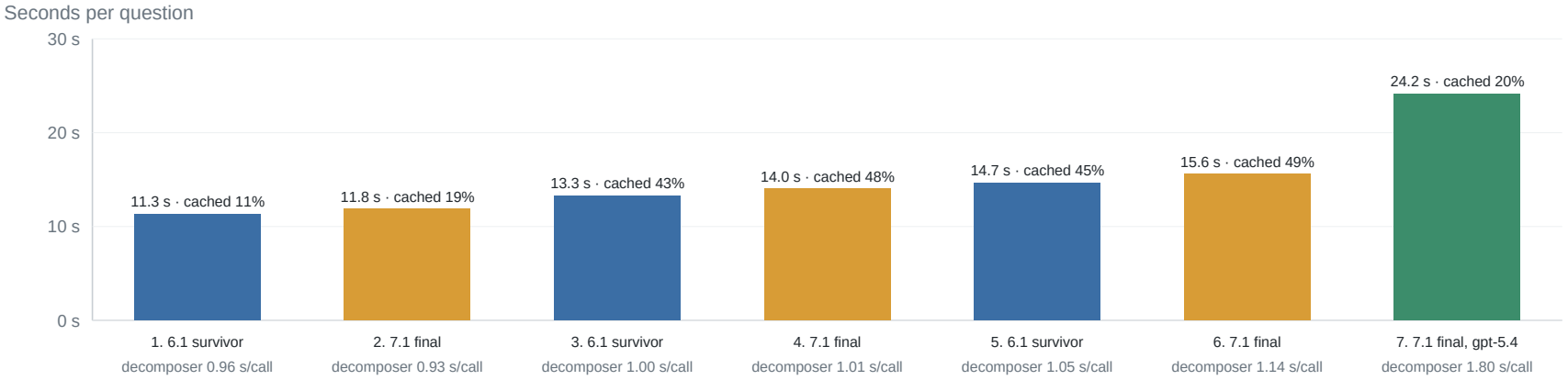
## 5. Tail and waste

| Configuration | List p50 | List p90 | List max | Max / mean | Seconds p50 | Seconds p90 | Seconds max | Spend on fails | Spend on partials | List $ per pass |
|---|---|---|---|---|---|---|---|---|---|---|
| 7.1 final | $0.0343 | $0.0902 | $0.2360 | 4.7x | 12.6 | 21.2 | 30.8 | 23% | 14% | $0.0773 |
| 6.1 survivor | $0.0333 | $0.0813 | $0.3144 | 6.5x | 11.8 | 20.1 | 30.5 | 13% | 34% | $0.0692 |
| 7.1 final, gpt-5.4 | $0.1120 | $0.3812 | $3.4416 | 10.2x | 22.3 | 35.6 | 51.1 | 6% | 53% | $0.4022 |

Most expensive question-runs:

| Configuration | Run | Question | Path | Verdict | List $ | Seconds | Calls |
|---|---|---|---|---|---|---|---|
| 7.1 final, gpt-5.4 | adoring_hopper | GT09 | Table | partial | $3.4416 | 42.0 | 12 |
| 7.1 final, gpt-5.4 | adoring_hopper | GT07 | Table | pass | $0.5770 | 51.1 | 15 |
| 7.1 final, gpt-5.4 | adoring_hopper | GT16 | Text, escalated | pass | $0.3322 | 22.3 | 5 |
| 6.1 survivor | hardcore_page | GT09 | Table | partial | $0.3144 | 20.4 | 9 |
| 7.1 final, gpt-5.4 | adoring_hopper | GT03 | Table | fail | $0.3036 | 32.5 | 8 |
| 7.1 final | confident_zuckerberg | GT09 | Table | pass | $0.2360 | 23.3 | 8 |
| 7.1 final, gpt-5.4 | adoring_hopper | GT05 | Table | pass | $0.2206 | 29.1 | 9 |
| 7.1 final, gpt-5.4 | adoring_hopper | GT17 | Text, ready at first assessment | pass | $0.2021 | 13.8 | 4 |
| 7.1 final, gpt-5.4 | adoring_hopper | GT14 | Table | pass | $0.1947 | 25.3 | 8 |
| 7.1 final, gpt-5.4 | adoring_hopper | GT12 | Table | pass | $0.1778 | 29.2 | 9 |

## 6. Comparability: run order, caching, and provider speed




| Run (order) | Results CSV | Finished | Quality | List $/q | Billed $/q | Billed / list | Cached | Seconds per q | Decomp. s/call | TOC s/call | Assessor s/call | Planner s/call | Answer s/call |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1. 6.1 survivor | eager_torvalds | 2026-10-04 01:41 | 0.737 | $0.0402 | $0.0368 | 0.92 | 11% | 11.3 | 0.96 | 1.80 | 1.58 | 0.93 | 1.30 |
| 2. 7.1 final | admiring_diffie | 2026-10-04 01:45 | 0.658 | $0.0507 | $0.0432 | 0.85 | 19% | 11.8 | 0.93 | 1.80 | 1.64 | 0.92 | 1.46 |
| 3. 6.1 survivor | hardcore_page | 2026-10-04 01:50 | 0.816 | $0.0570 | $0.0372 | 0.65 | 43% | 13.3 | 1.00 | 1.76 | 1.82 | 0.94 | 1.32 |
| 4. 7.1 final | eloquent_hoare | 2026-10-04 01:55 | 0.711 | $0.0459 | $0.0286 | 0.62 | 48% | 14.0 | 1.01 | 2.13 | 1.91 | 0.98 | 1.52 |
| 5. 6.1 survivor | confident_stallman | 2026-10-04 02:00 | 0.816 | $0.0485 | $0.0310 | 0.64 | 45% | 14.7 | 1.05 | 2.41 | 2.13 | 1.09 | 1.64 |
| 6. 7.1 final | confident_zuckerberg | 2026-10-04 02:06 | 0.763 | $0.0539 | $0.0325 | 0.60 | 49% | 15.6 | 1.14 | 2.52 | 2.08 | 1.10 | 1.63 |
| 7. 7.1 final, gpt-5.4 | adoring_hopper | 2026-10-04 10:12 | 0.868 | $0.3387 | $0.2787 | 0.82 | 20% | 24.2 | 1.80 | 4.42 | 4.38 | 1.34 | 2.35 |

*Run times from: run_manifest.csv.*

## Warnings

- Single run: 7.1 final, gpt-5.4. One run cannot show a difference smaller than the run-to-run spread of a repeated configuration.
- 6.1 survivor: list cost per question ranges from $0.0402 to $0.0570 across its own runs (42% apart).
- Cached share ranges from 11% to 49% across runs (it grows with repeats of the same questions). Billed cost depends on run order; compare list cost.

- The decomposer's median call took 0.93-1.80 s across runs (93% apart). Its prompt is the same in every configuration, so time differences of this size are the provider's speed, not the system.

## 7. Safeguard markers

| Configuration | Question-runs | Evidence notice | Noticed: pass / partial / fail | Grounding checked | Flagged | Flagged: pass / partial / fail | Check passed, judge failed | Input sanitized |
|---|---|---|---|---|---|---|---|---|
| 7.1 final | 57 | 17 (30%) | 11/4/2 | 57 | 10 (18%) | 8/1/1 | 12 | 0 |
| 6.1 survivor | 57 | 17 (30%) | 12/4/1 | not checked | - | - | - | 0 |
| 7.1 final, gpt-5.4 | 19 | 5 (26%) | 4/1/0 | 19 | 1 (5%) | 1/0/0 | 2 | 0 |

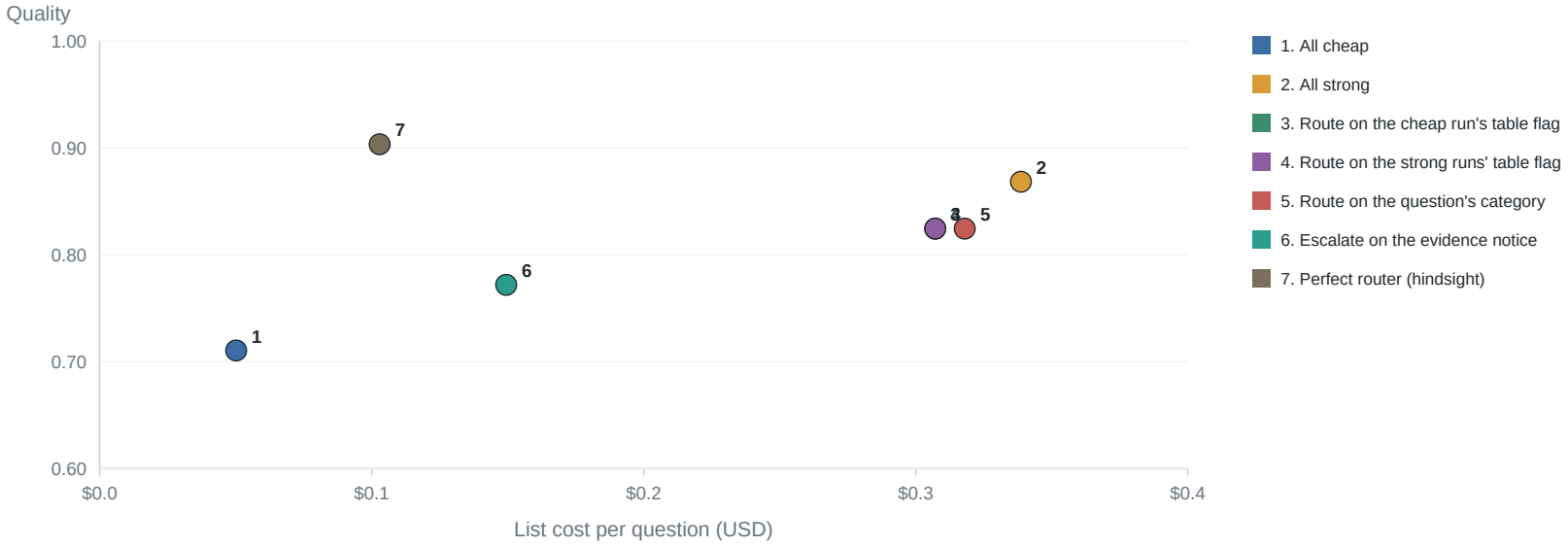
*A flagged answer the judge passed is a false alarm the user still sees (policy 'surface' labels it). 'Check passed, judge failed' counts wrong answers the deterministic check cannot see.*

Spans the grounding check could not find in the evidence:

| Span | Kind | Times | Judge: pass / partial / fail | Configurations |
|---|---|---|---|---|
| Russia-related Olympic | name | 3 | 3/0/0 | 7.1 final |
| Russian Empire | name | 3 | 3/0/0 | 7.1 final; 7.1 final, gpt-5.4 |
| Winter Olympic Games Sport | name | 2 | 2/0/0 | 7.1 final |
| Spanish armies, led by Fernando Álvarez de Toledo, 3rd Duke of Alba, defending the claim of King Philip II of Spain to the Portuguese throne, defeat the armies of Portuguese claimant António, Prior of Crato | quotation | 1 | 1/0/0 | 7.1 final |
| Summer Olympic Games Sport | name | 1 | 0/0/1 | 7.1 final |
| To Barack Obama | name | 1 | 0/1/0 | 7.1 final |
| To Bill Clinton | name | 1 | 0/1/0 | 7.1 final |

## 8. Routing

Cheap: 7.1 final. Strong: 7.1 final, gpt-5.4. Text categories for the category policy: Direct fact; Direct fact and quotation; Process explanation; Inference limitation; Diagnostic reasoning and limitations; People connection: intermediary person; People connection: shared work and roles.




| Policy | Rule | Quality | List $/q | Seconds/q | Sent to strong | Paid twice | Saving vs all strong |
|---|---|---|---|---|---|---|---|
| 1. All cheap | every question on the cheap configuration | 0.711 | $0.0502 | 13.8 | 0% | 0% | 85% |
| 2. All strong | every question on the strong configuration | 0.868 | $0.3387 | 24.2 | 100% | 0% | 0% |
| 3. Route on the cheap run's table flag | strong when the cheap run's TOC review flagged a table | 0.825 | $0.3072 | 22.6 | 74% | 0% | 9% |
| 4. Route on the strong runs' table flag | strong when most strong runs flagged a table for the question | 0.825 | $0.3072 | 22.6 | 74% | 0% | 9% |
| 5. Route on the question's category | strong unless the category is one of the text categories | 0.825 | $0.3180 | 23.0 | 79% | 0% | 6% |
| 6. Escalate on the evidence notice | cheap first; rerun on strong when the evidence notice fired (both paid) | 0.772 | $0.1494 | 22.5 | 30% | 30% | 56% |
| 7. Perfect router (hindsight) | strong only where the strong mean scores higher; the lower bound for any router | 0.904 | $0.1029 | 17.9 | 25% | 0% | 70% |

*Simulated on saved runs: a routed question takes the strong configuration's mean outcome, cost, and time for that question. A routed system is a different run, so the simulation predicts and a live run confirms.*

*Route mix: no run on this set records a router decision (column 'route').*